

PAPER_696: Antennae Galaxies (NGC 4038/4039): Active Merger and Shock-Triggered Star Formation

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Class: AntennaeMergerNGC4038NGC4039

CP4 Entry: #280

Keywords: Antennae, NGC 4038, NGC 4039, galaxy merger, dynamical friction, shock SFR, UQFF

Session: 174 | **Version:** v5.31

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Abstract

The Antennae Galaxies (NGC 4038/4039) are the closest major galaxy merger, exhibiting spectacular tidal tails, active starburst regions, and X-ray bright shock ionization zones. UQFF modifies the merger gravity and shock-enhanced star formation.

1. Parameters

Parameter	Value
M_{4038}	$10^{11} M_{\odot}$
M_{4039}	$8 \times 10^{10} M_{\odot}$
Separation	7 kpc
SFR (burst)	$20 M_{\odot}/\text{yr}$

2. UQFF Merger Gravity

$$g_{\text{merge}} = \frac{G(M_1 + M_2)}{r^2} (1 + f_{\text{TRZ}}) + \rho_{\text{shock}} V g \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}}$$

3. Dynamical Friction (Chandrasekhar)

$$t_{\text{df}} = \frac{1.17 v^3}{G^2 M_{\text{tot}} \rho_{\text{avg}} \ln \Lambda}$$

4. UQFF Star Formation Rate

$$\text{SFR}_{\text{UQFF}} = \text{SFR}_{\text{burst}} \frac{\rho_{\text{shock}}}{\rho_{\text{UA}}} \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} \right)$$

5. Binding Energy

$$E_{\text{bind}} = -\frac{GM_1 M_2}{d_{\text{sep}}}$$

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Hibbard et al. (2001), AJ, 122, 2969
- UQFF Framework, Star-Magic Session 174

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